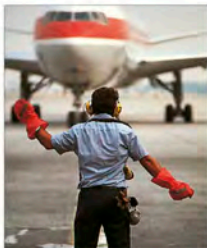


SOUND



Astronauts talk to each other by radio because their voices can't travel through empty space. Sound travels in waves. When these waves move through air, the air molecules move quickly,

or vibrate. If there are no air molecules, no sound is made, because there is nothing to vibrate. When you shout, the vocal cords in your throat vibrate. These vibrations pass through your mouth into the air, making the air itself vibrate. Your ear picks up the vibrations and you can hear them as sound.



Deafening decibels

The loudness of sounds is measured in decibels. When airplanes land, they reach a very high number of decibels. This ground controller is wearing ear protectors to prevent being deafened by the noise.



Wailing whales

Whales communicate under water over huge distances. The sound of a great whale may travel hundreds of miles under water. Sound travels faster and farther through liquids and solids than through air. This is because molecules in liquids and solids are more tightly packed.

The bottle tops clink together when the jingling stick is shaken.

Lentils inside the shaker make a soft, rattling sound.

Strings vibrate as the instrument is plucked. The tighter the string, the higher the note.

